

```

root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400GP', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGP_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
if exist('used_ds400GP', 'var')
    used_ds = used_ds400GP;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = {cuts2fit400GP.cuts_off110f};
    sel_info.en_range = 50:10:150;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400off110_2D';
    sel_info.bg_par = {};
    used_ds('Ei400Br110dir111') = sel_info;
    %
    sel_info.cuts = {cuts2fit400GP.cuts_off200f};
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400off200_2D';
    sel_info.bg_par = {};
    used_ds('Ei400Br200dir111') = sel_info;

    used_ds400GP = used_ds;
end

```

```
used_ds =
```

Map with properties:

```

    Count: 0
    KeyType: char
    ValueType: any

```

Show all accessible properties of Map

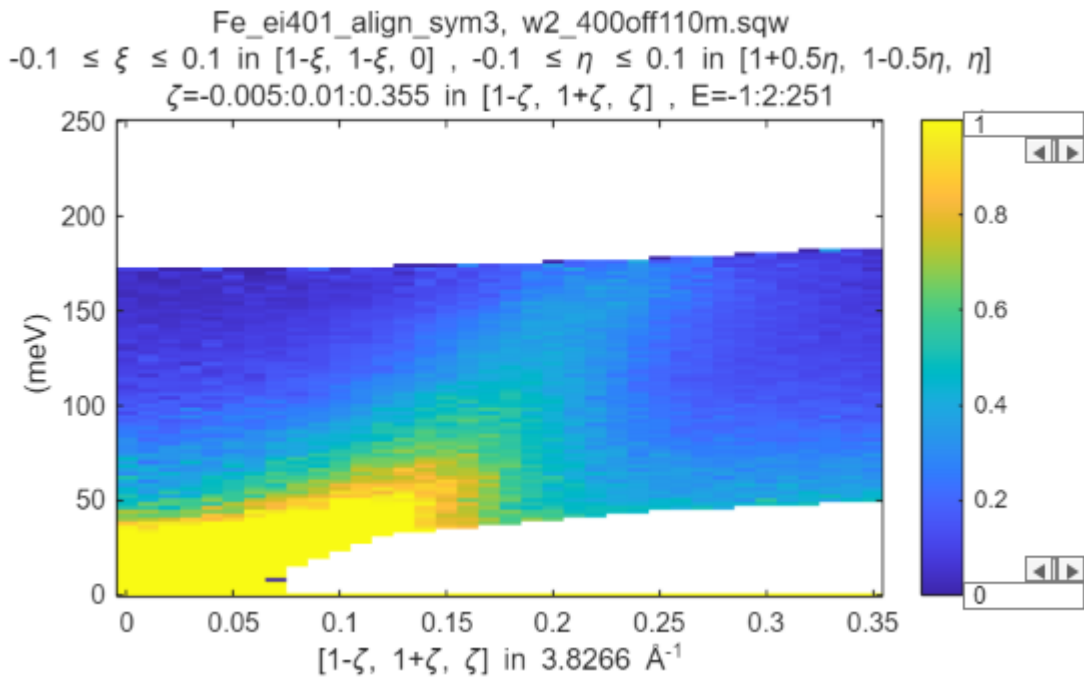
```

select = 'Ei400Br110dir111';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400, 600, 'S');
sample = IX_sample(true, [1, 0, 0], [0, 1, 0], 'cuboid', [0.04, 0.03, 0.02]);
for i = 1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



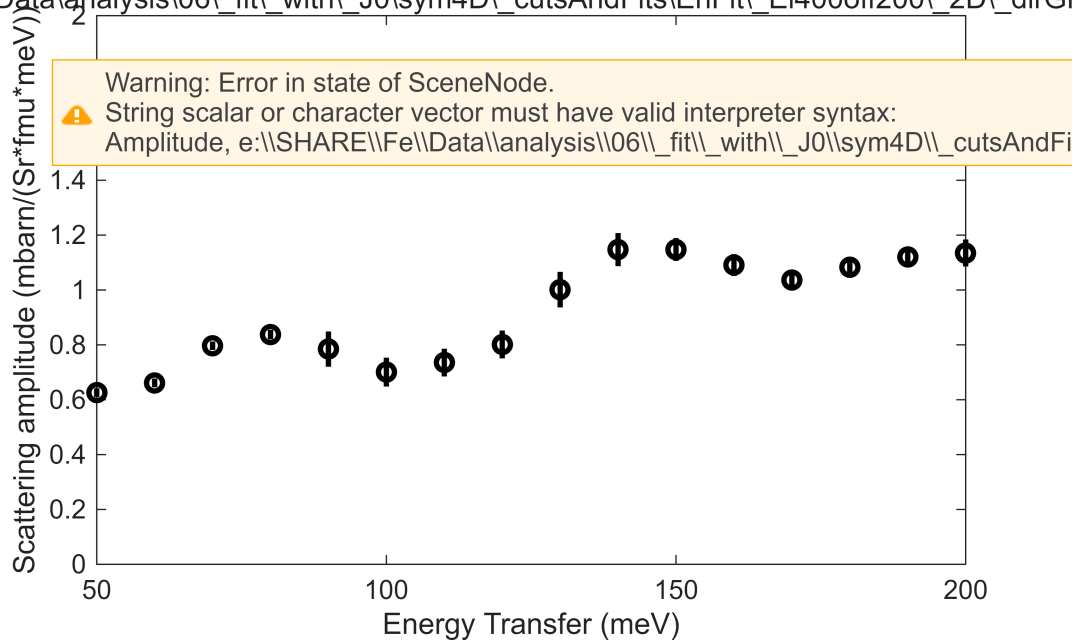
```
dir_name = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);
cut_dimensions = sel_info.cut_dim;

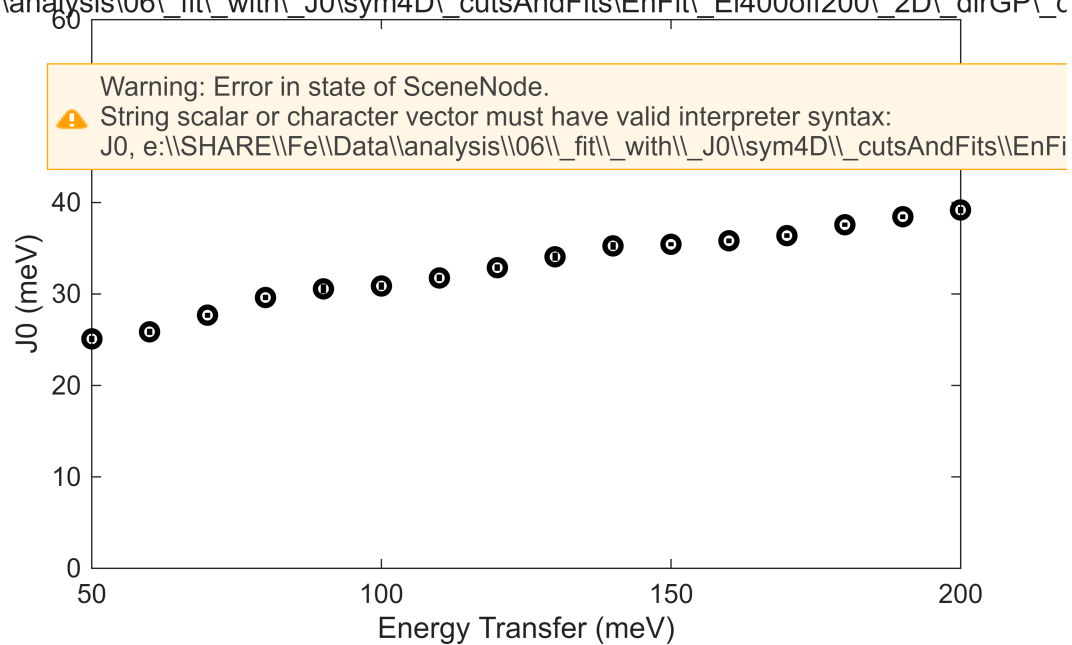
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

```
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

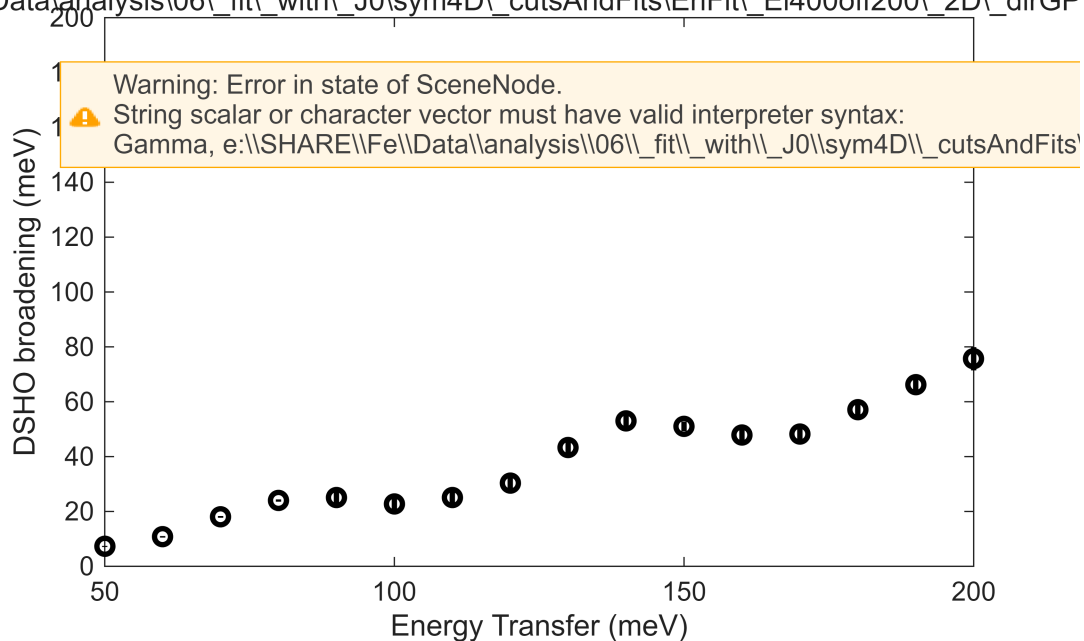
Fe\\Data\\analysis\\06_fit_with_J0\\sym4D_cutsAndFits\\EnFit_Ei400off200_2D_dirGl



Data\\analysis\\06_fit_with_J0\\sym4D_cutsAndFits\\EnFit_Ei400off200_2D_dirGP_c



\\e\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei400off200_2D_dirGP



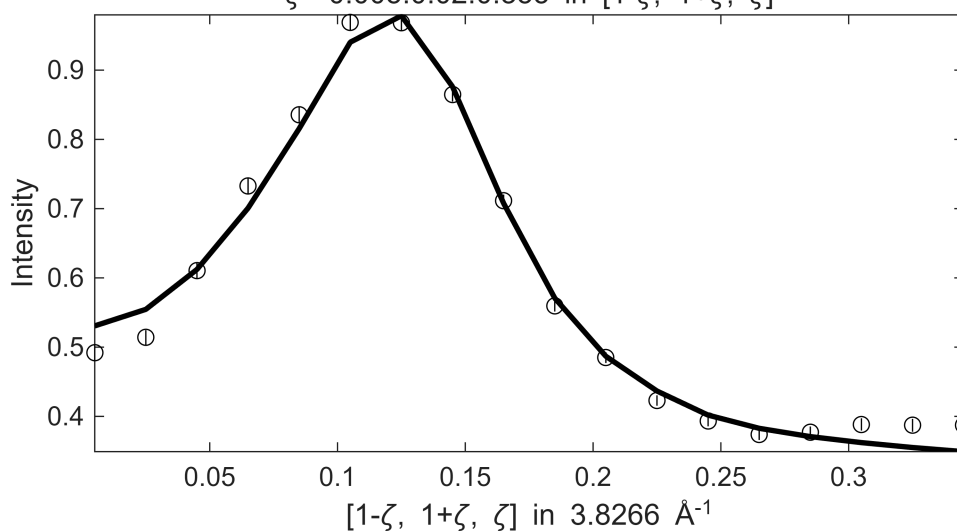
**** En = 50±10

Fe_ei401_align_sym3, w2_400off110m.sqw

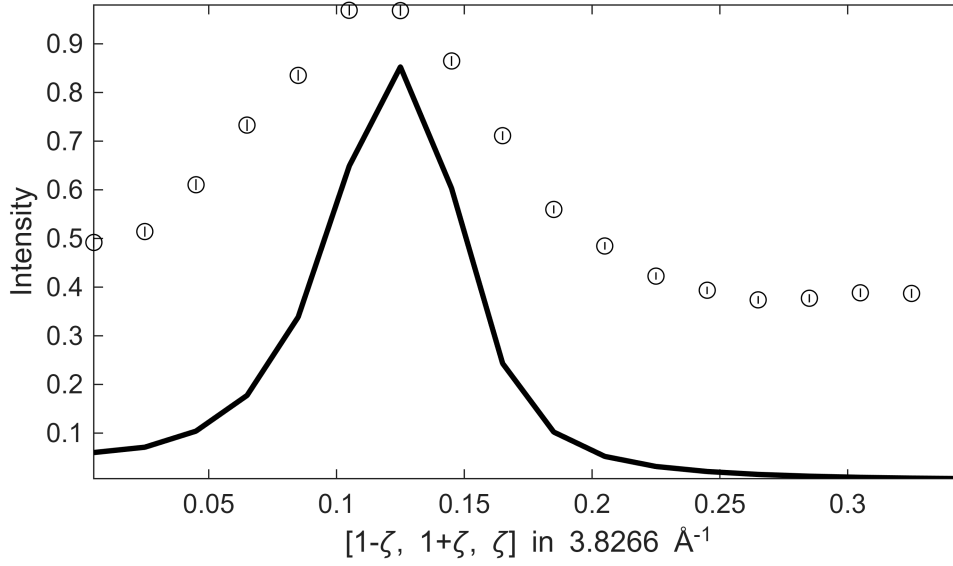
S=0.79±0.035; J0= 25±0.24; gamma=12.1352±1.18874

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 40 ≤ E

ζ=-0.005:0.02:0.355 in [1-ζ, 1+ζ, ζ]

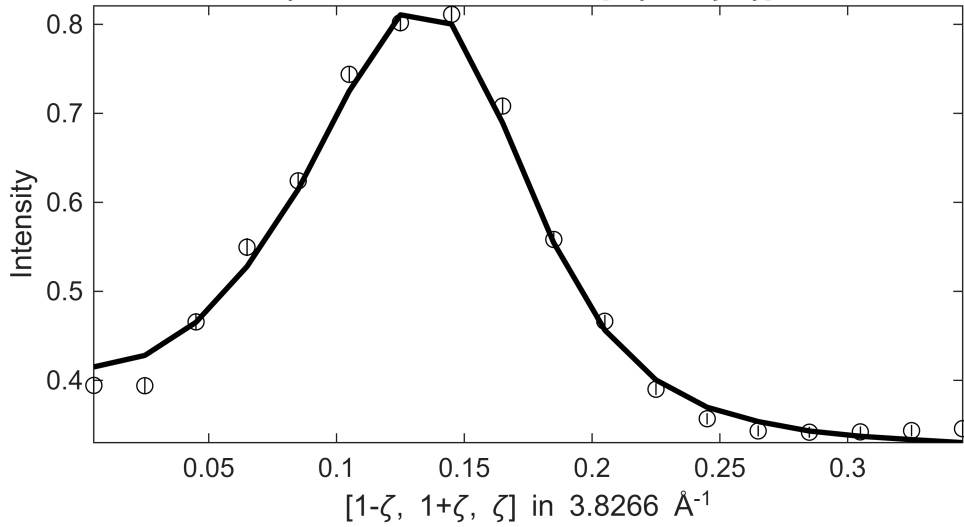


Fe_ei401_align_sym3, w2_400off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $40 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

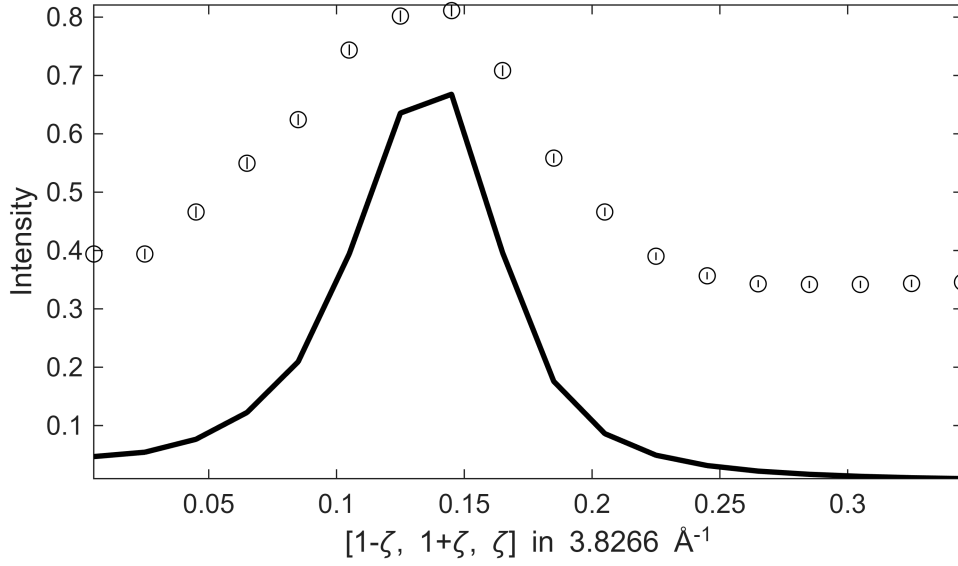


 **** En = 60 ± 10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.82 \pm 0.034$; $J_0= 26 \pm 0.25$; $\gamma=17.2849 \pm 1.39451$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

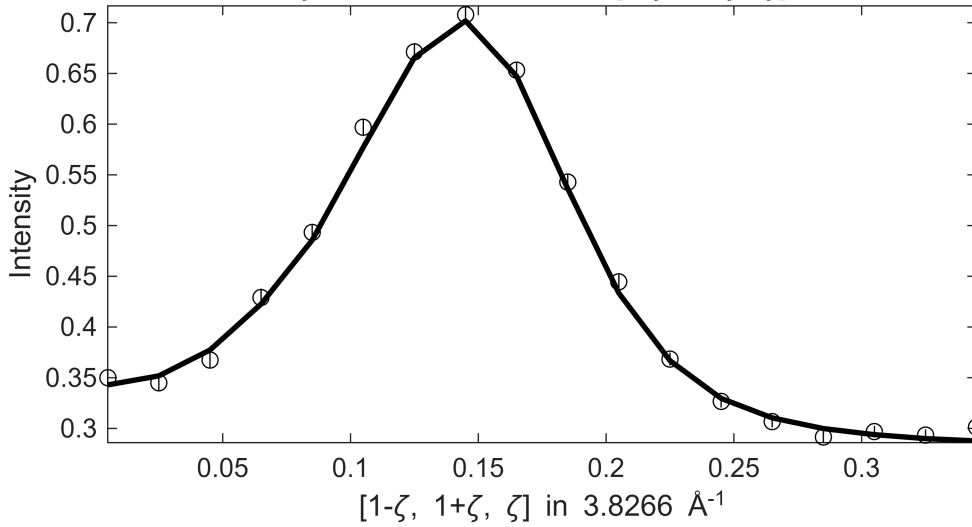


Fe_ei401_align_sym3, w2_400off110m.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $50 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



 **** $E_n = 70 \pm 10$

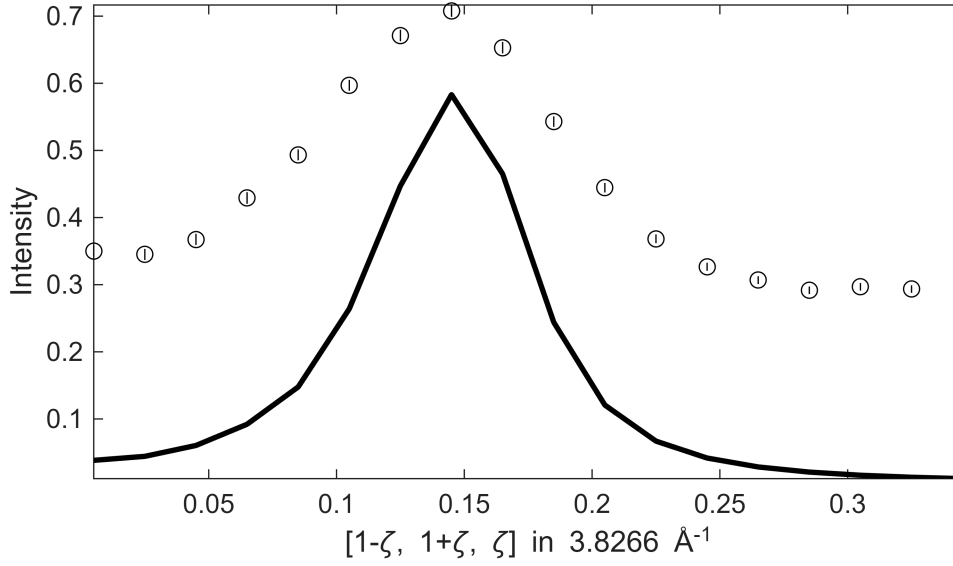
Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.81 \pm 0.011$; $J_0 = 28 \pm 0.19$; $\gamma = 21.5803 \pm 0.238509$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off110m.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq E$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



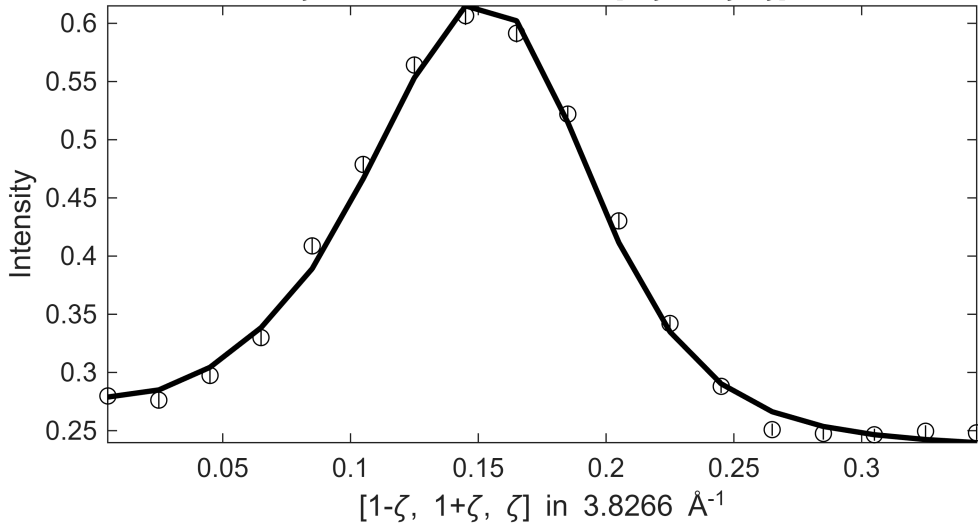
**** En = 80 ± 10

Fe_ei401_align_sym3, w2_400off110m.sqw

$S=0.81 \pm 0.011$; $J_0=29 \pm 0.17$; $\gamma=24.434 \pm 0.24912$

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq E$

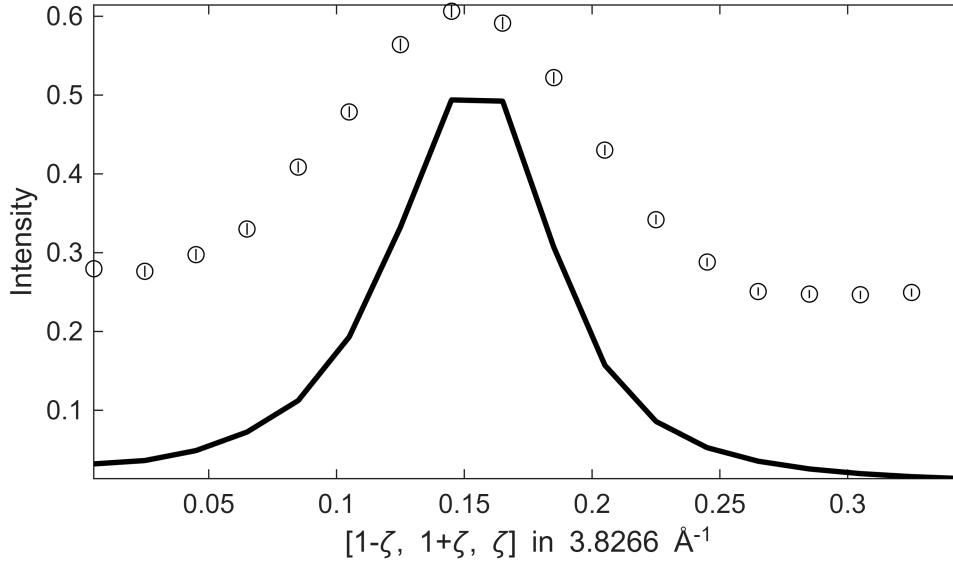
$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off110m.sqw

$.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq E$

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



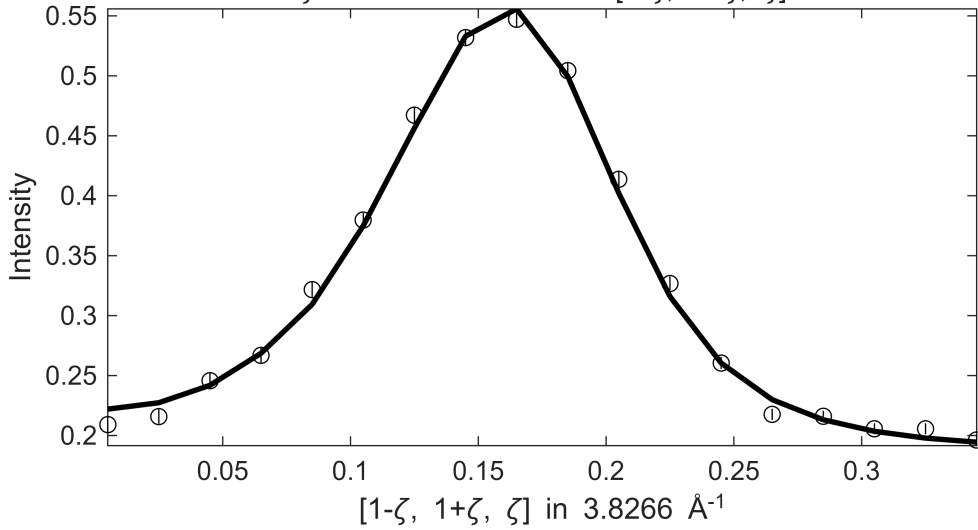
**** En = 90±10

Fe_ei401_align_sym3, w2_400off110m.sqw

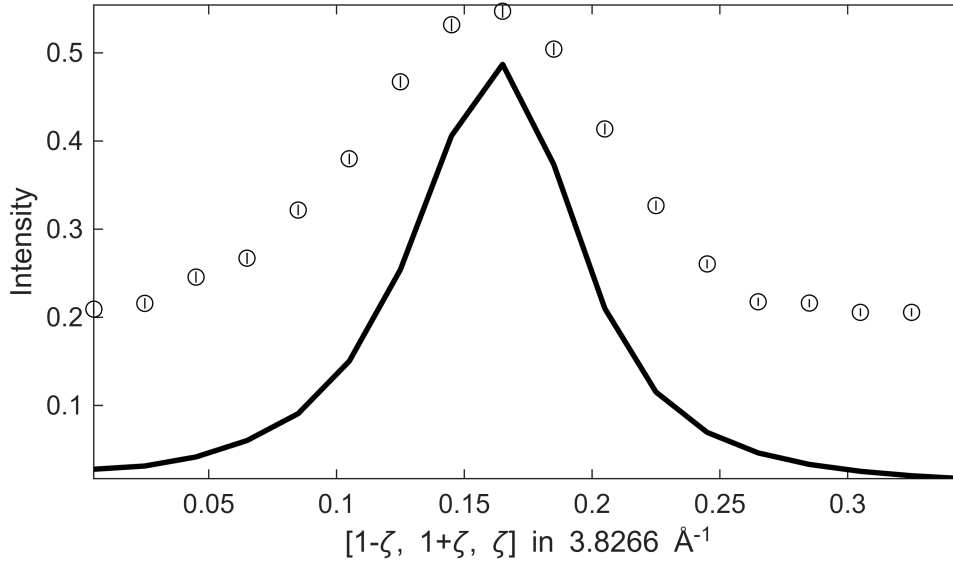
S=0.84±0.012; J0= 30±0.16; gamma=27.9743±0.282665

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$:

$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

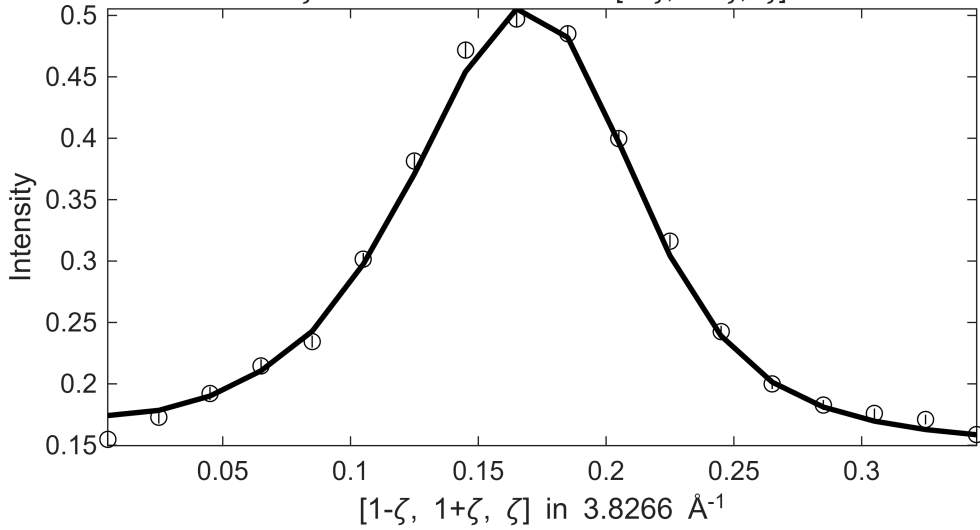


Fe_ei401_align_sym3, w2_400off110m.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$:
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

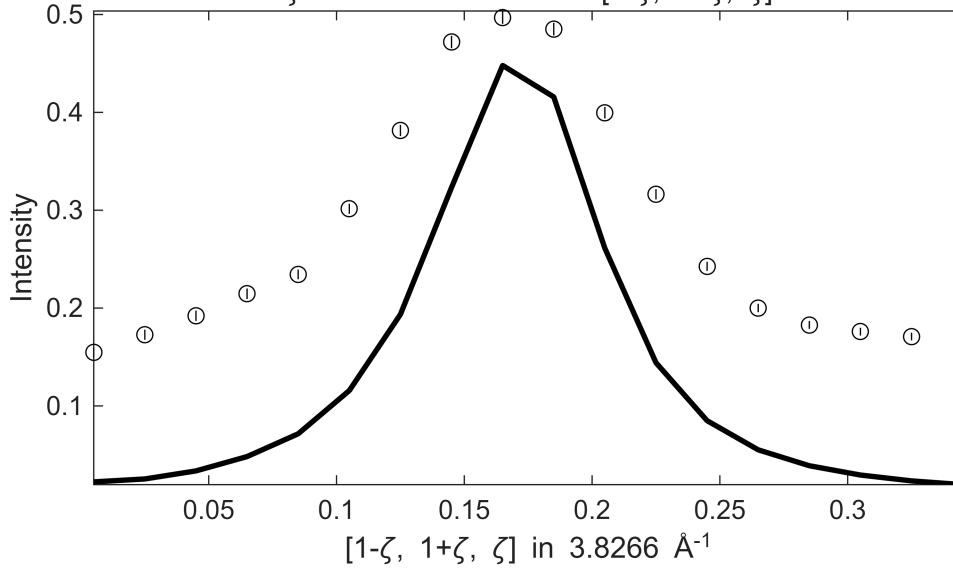


 **** En = 100±10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.85\pm0.013$; $J0= 31\pm0.15$; $\gamma=29.8476\pm0.338152$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$:
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

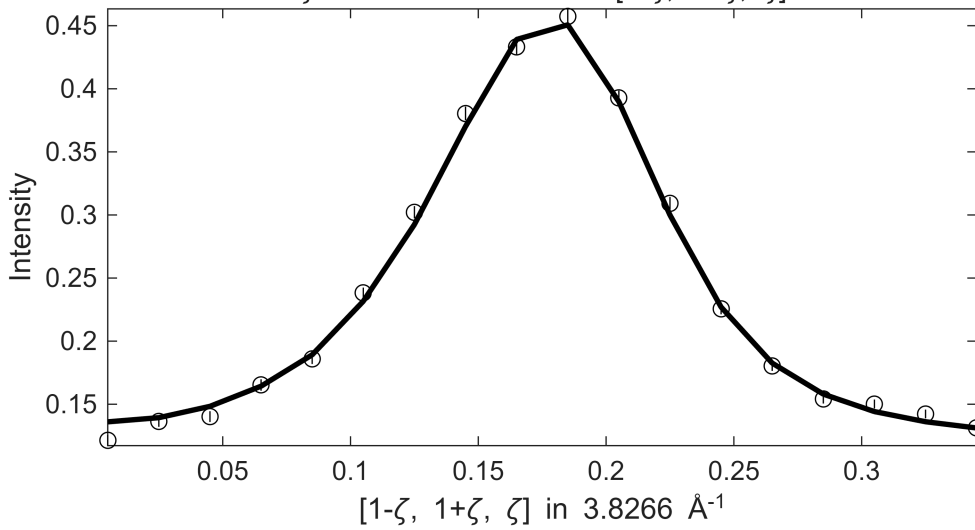


Fe_ei401_align_sym3, w2_400off110m.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$:
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

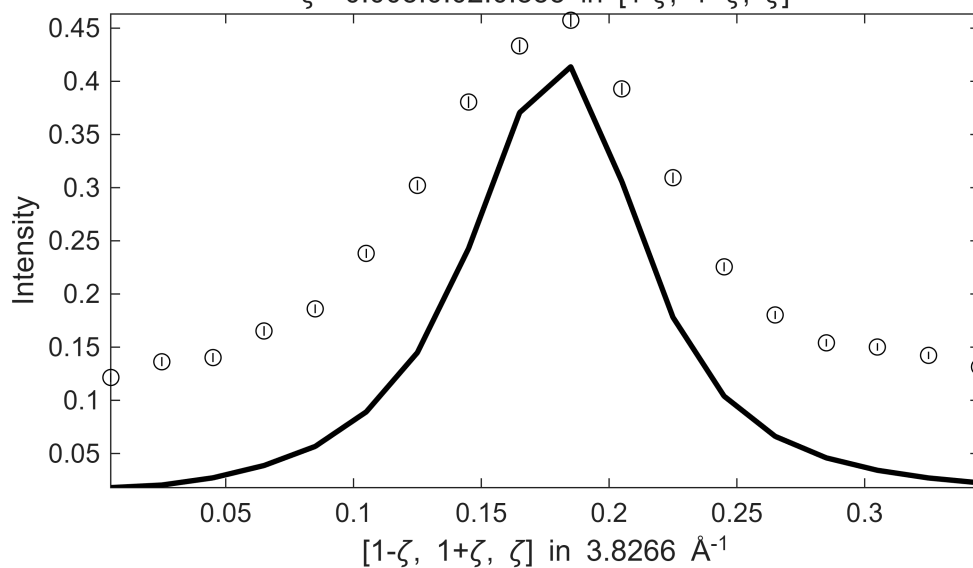


 **** En = 110±10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.83 \pm 0.013$; $J_0 = 32 \pm 0.13$; $\gamma = 32.3812 \pm 0.389865$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$:
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

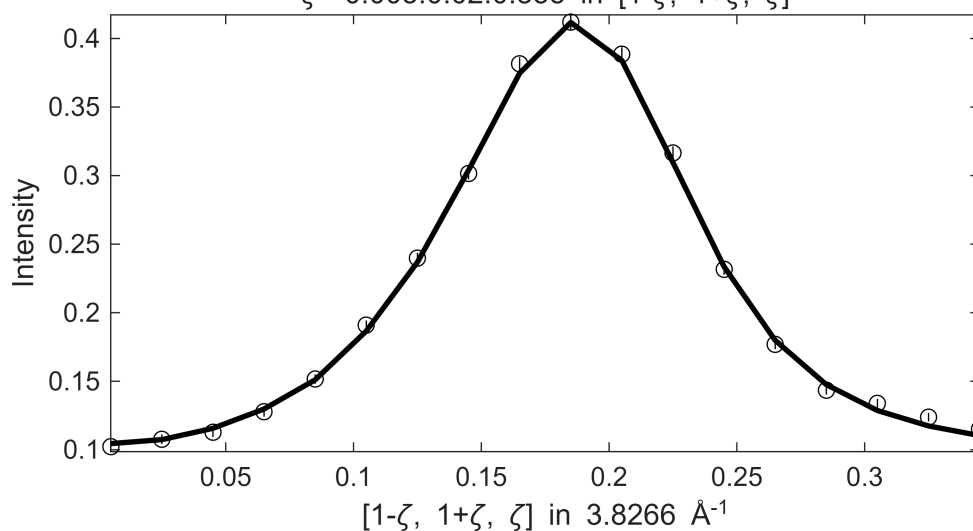


Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

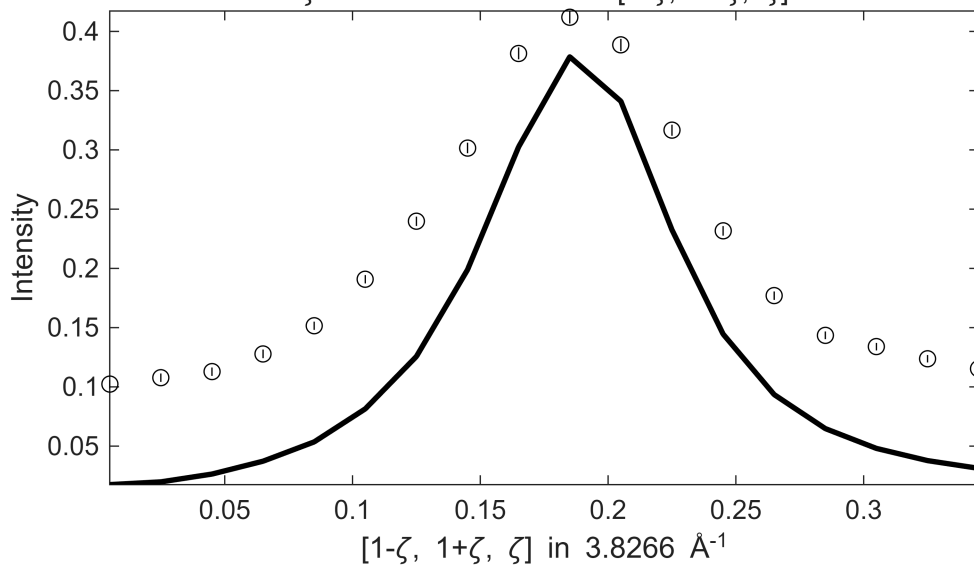


**** En = 120 ± 10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.91 \pm 0.017$; $J_0 = 33 \pm 0.12$; $\gamma = 40.0857 \pm 0.595541$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $110 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

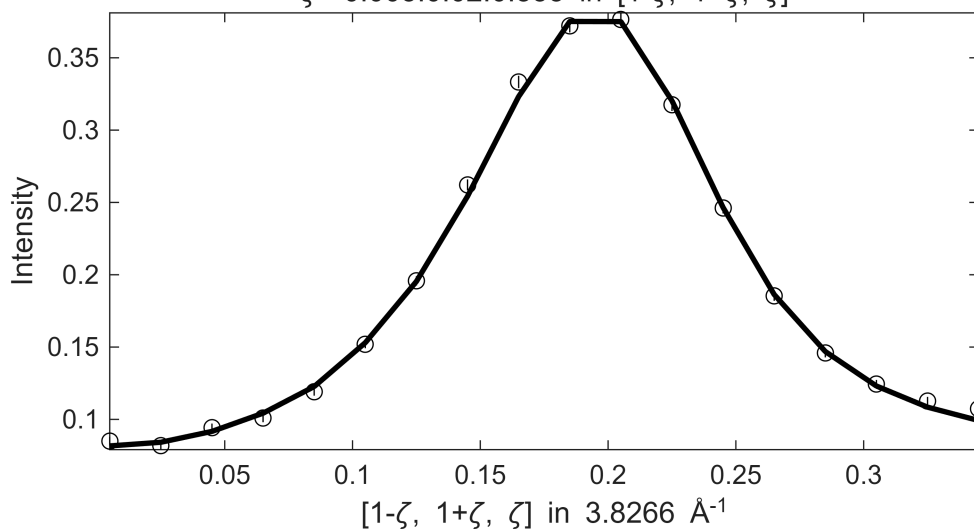


Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $110 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



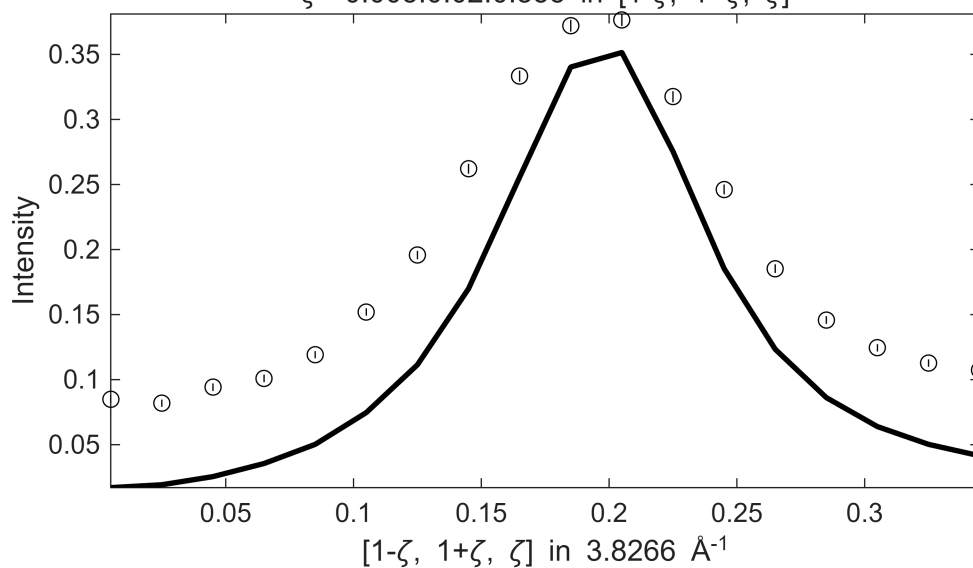
 **** En = 130±10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.99\pm0.029$; $J_0=34\pm0.18$; $\gamma=46.4188\pm1.69416$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $120 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off110m.sqw

$|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $120 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

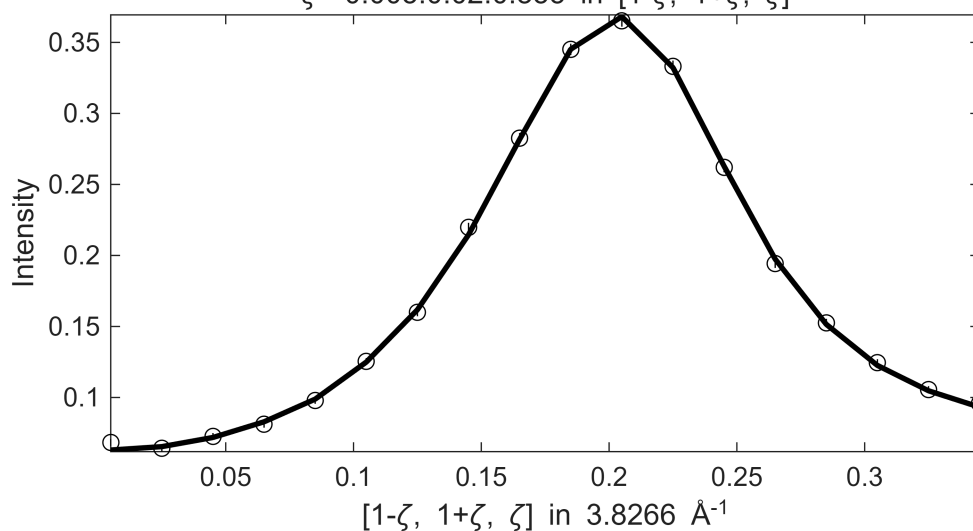


**** En = 140±10

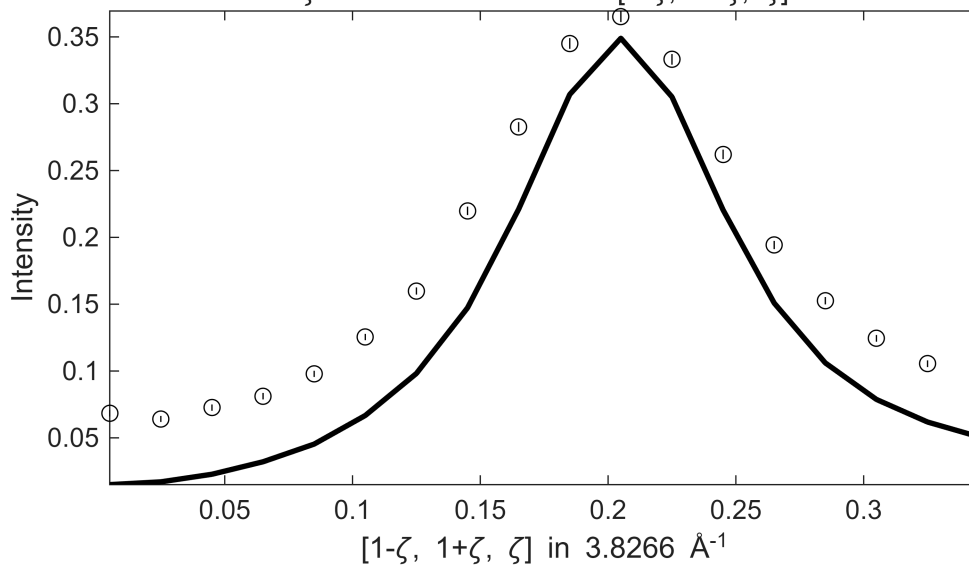
Fe_ei401_align_sym3, w2_400off110m.sqw

S= 1±0.021; J0= 35±0.11; gamma=50.1866±1.10183

$|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $130 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

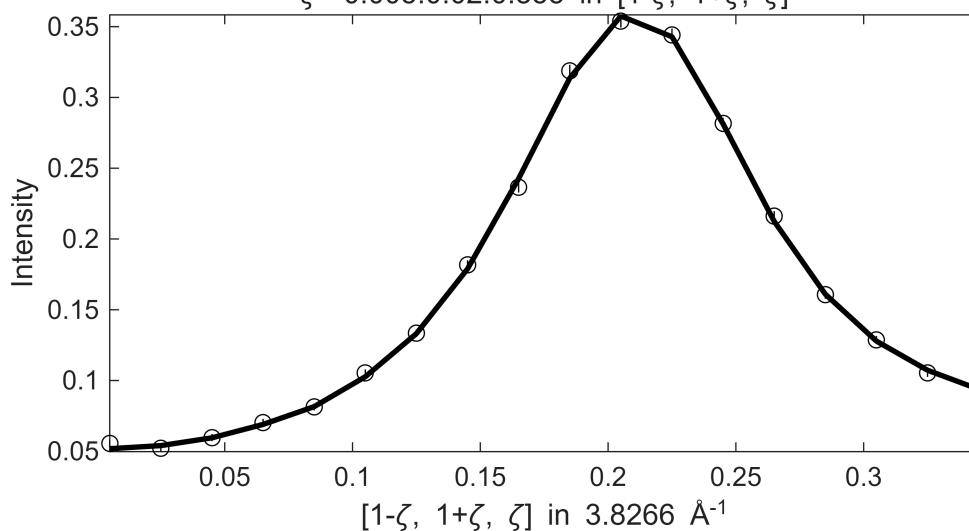


Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $130 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

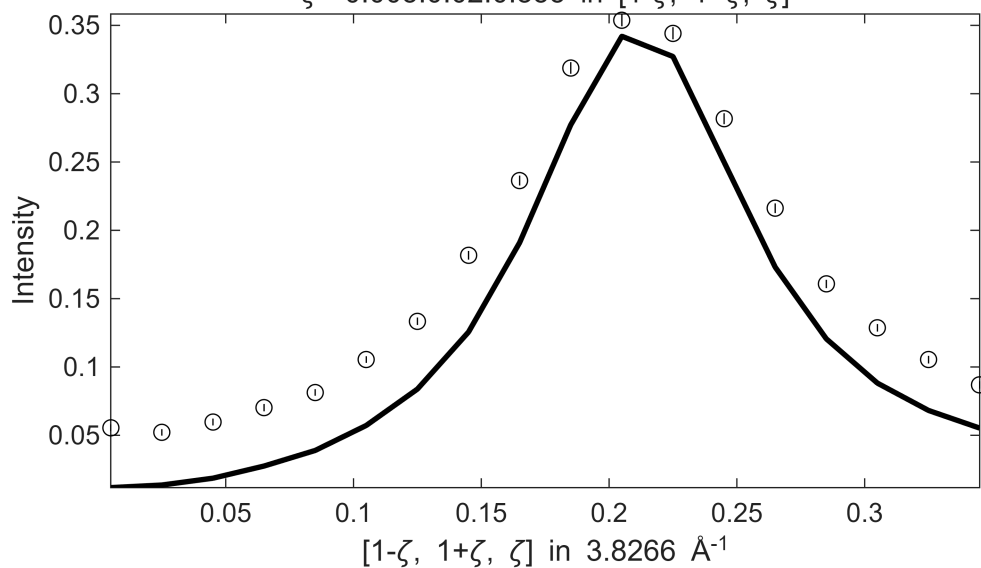


**** En = 150±10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S = 1 \pm 0.022$; $J_0 = 35 \pm 0.13$; $\gamma = 49.6914 \pm 1.23382$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $140 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

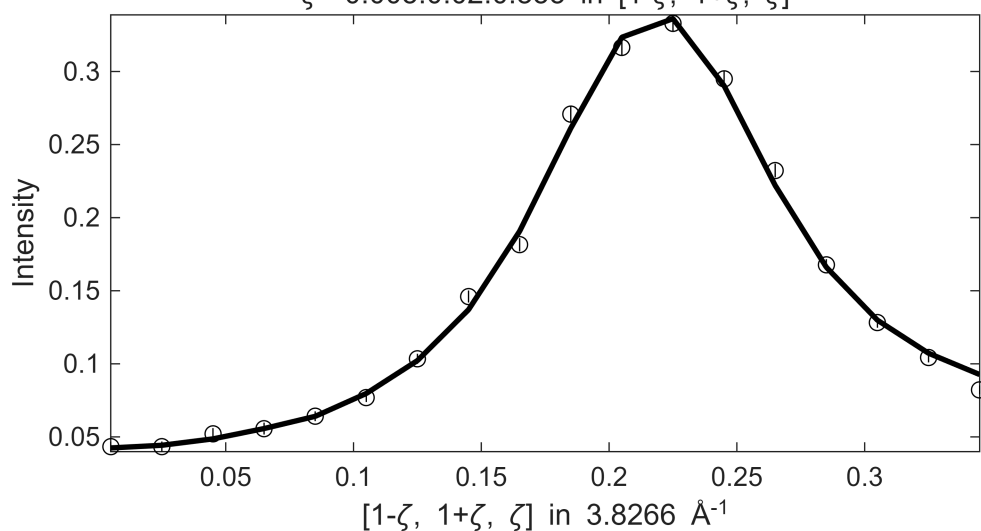


Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $140 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

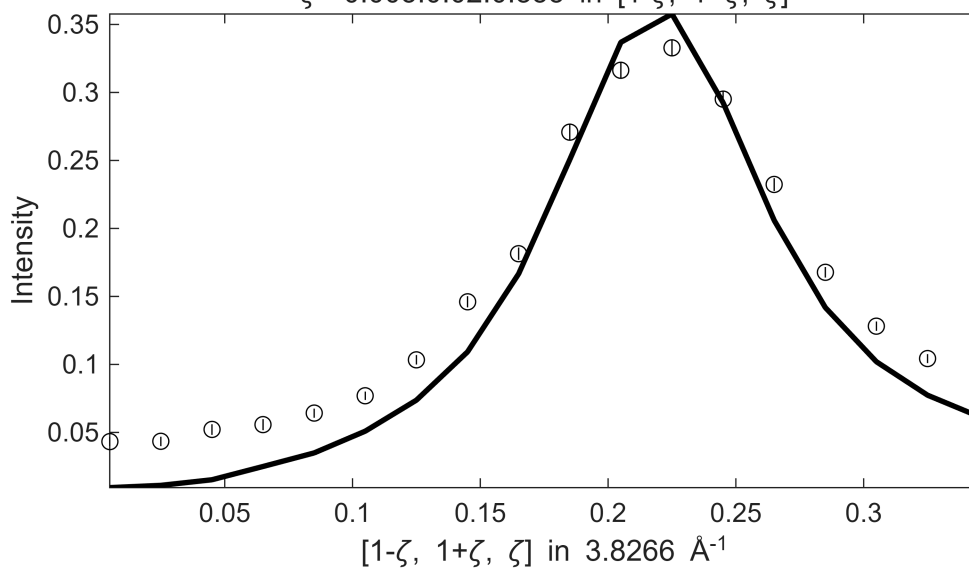


**** En = 160±10

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.99\pm0.027$; $J0= 35\pm0.17$; $\gamma=48.0312\pm1.59187$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $150 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



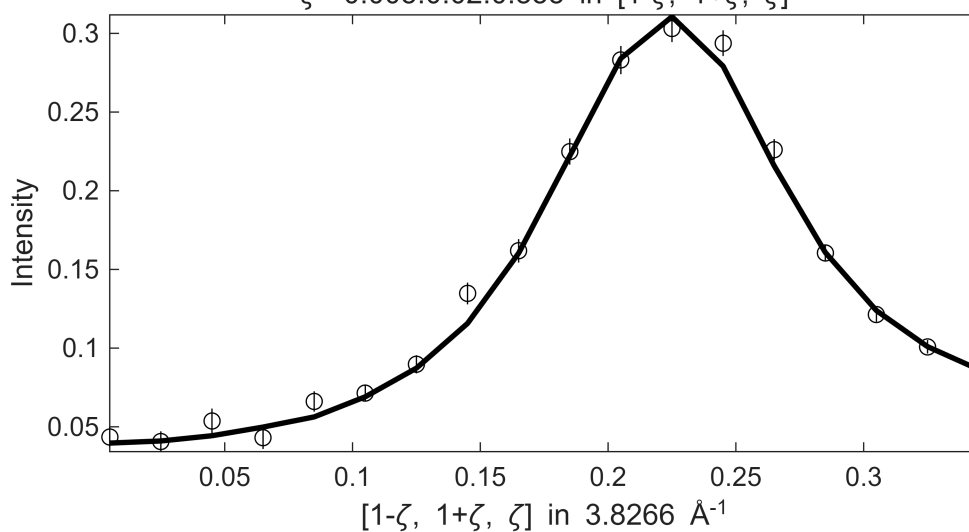
Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $150 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



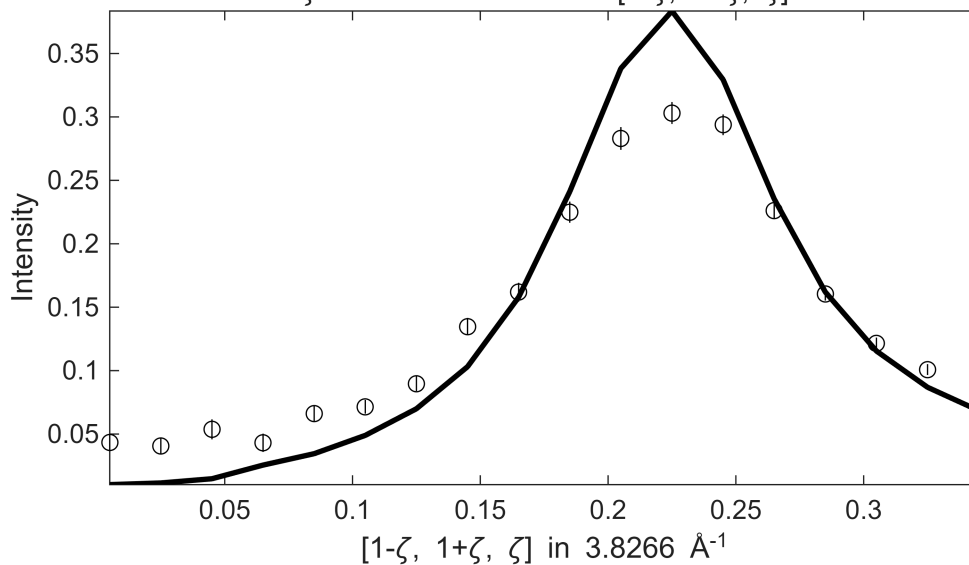
**** En = 170±10

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.5253

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.91\pm0.041$; $J_0=35\pm0.25$; $\gamma=44.7963\pm2.40881$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $160 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



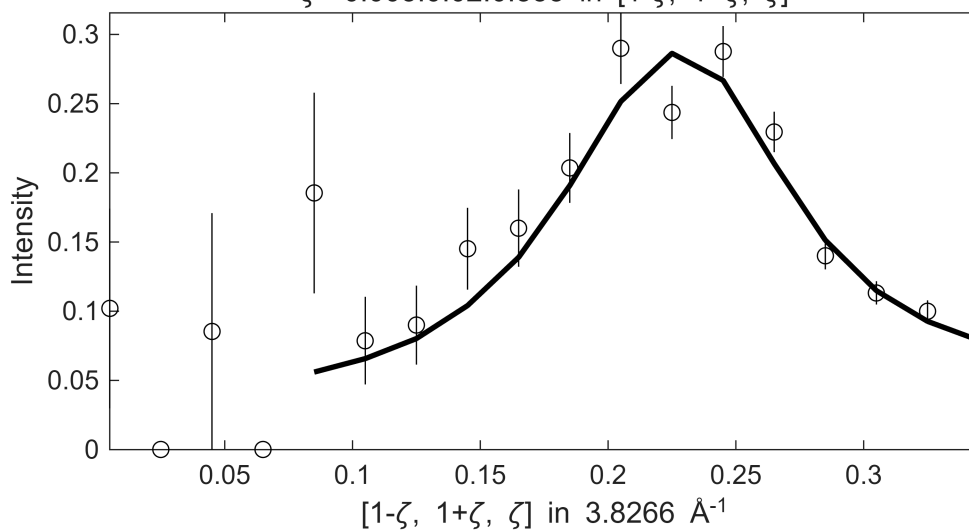
Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $160 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



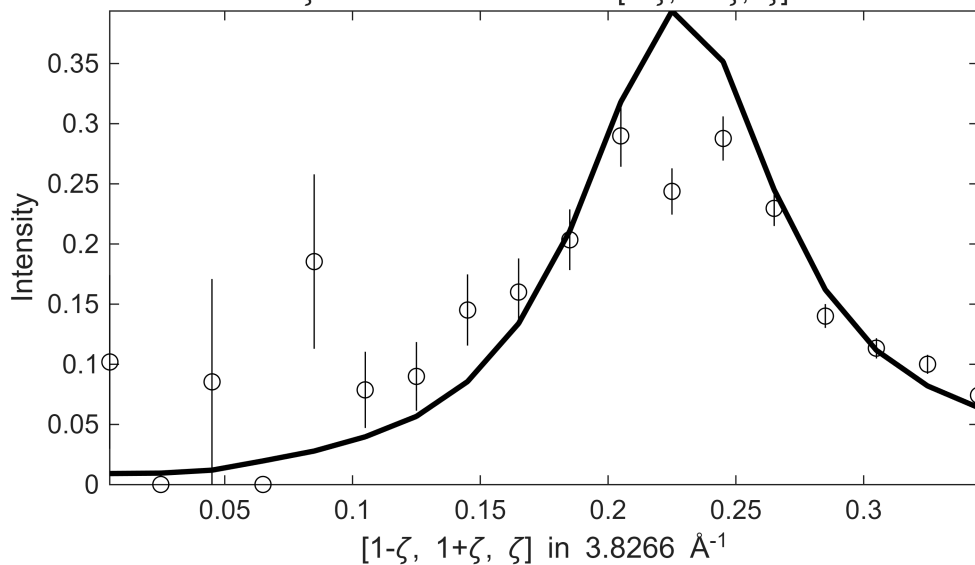
**** En = 180 ± 10

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.5253

Fe_ei401_align_sym3, w2_400off110m.sqw
 $S=0.74 \pm 0.091$; $J_0=34 \pm 0.44$; $\gamma=36.7514 \pm 4.71186$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $170 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



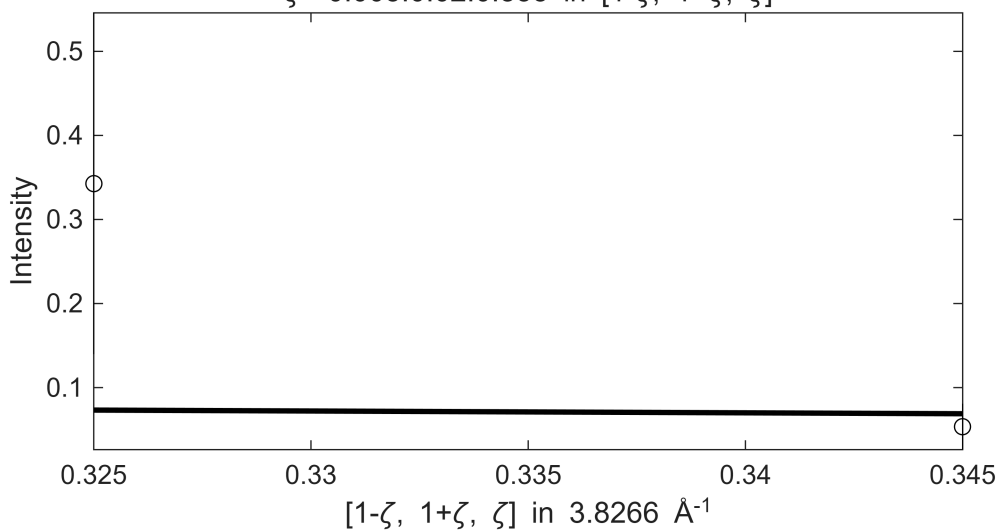
Fe_ei401_align_sym3, w2_400off110m.sqw
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $170 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



**** En = 190±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.50505

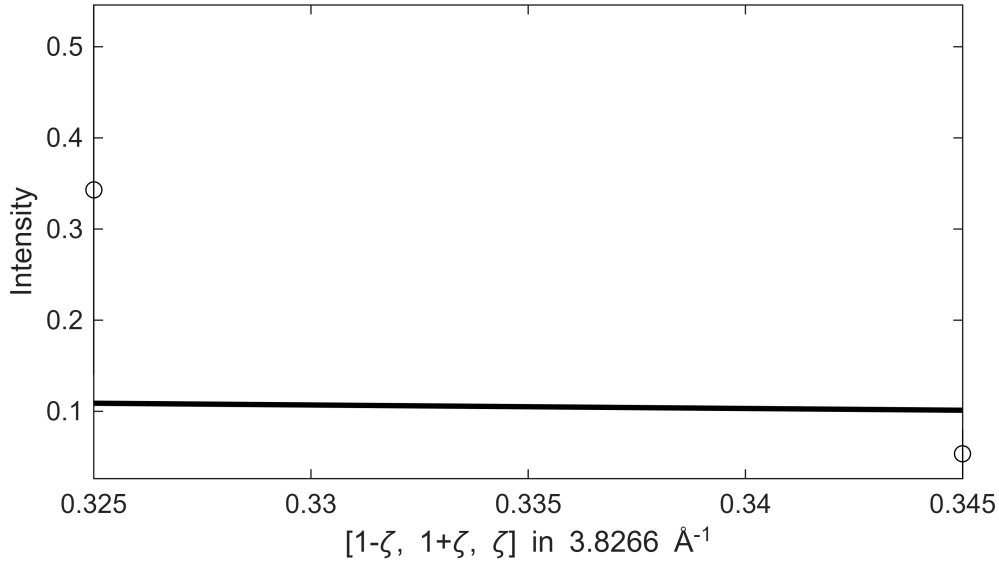
Fe_ei401_align_sym3, w2_400off110m.sqw
 $S = 1.5 \pm 6.1$; $J_0 = 43 \pm 39$; $\gamma = 121.819 \pm 609.134$
 $|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $180 \leq E$
 $\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



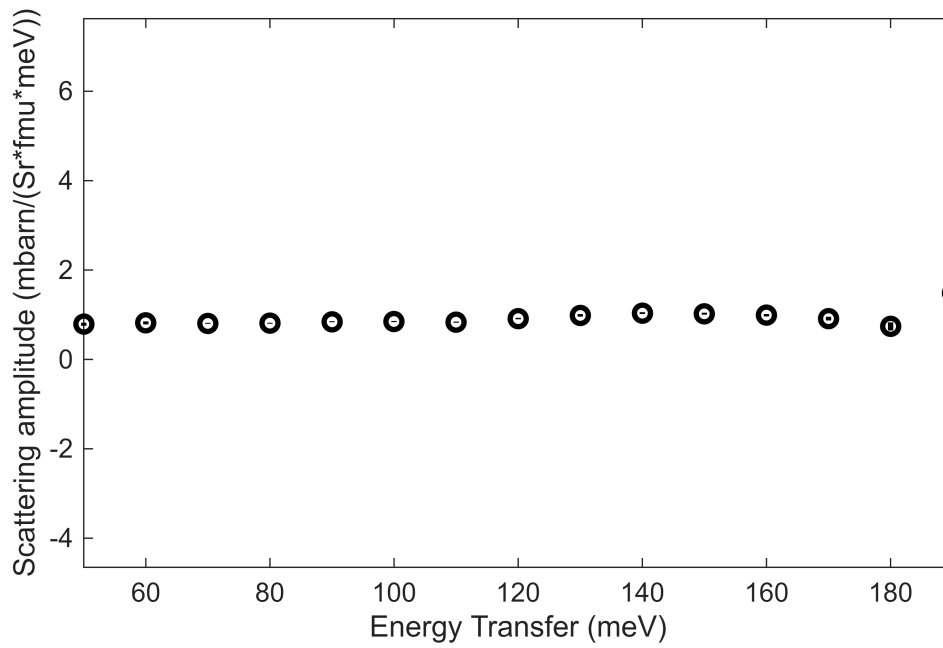
Fe_ei401_align_sym3, w2_400off110m.sqw

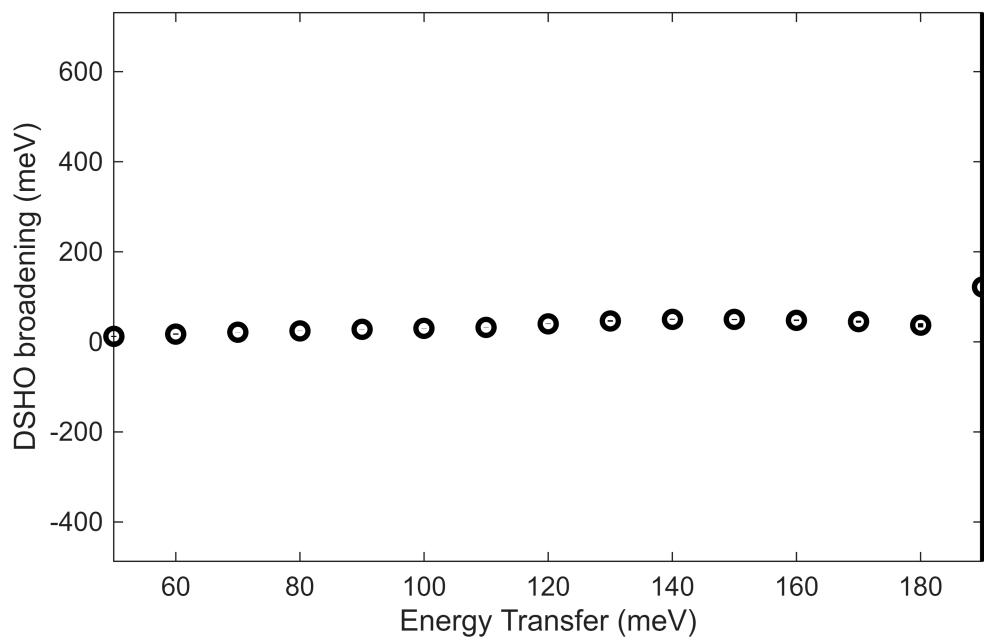
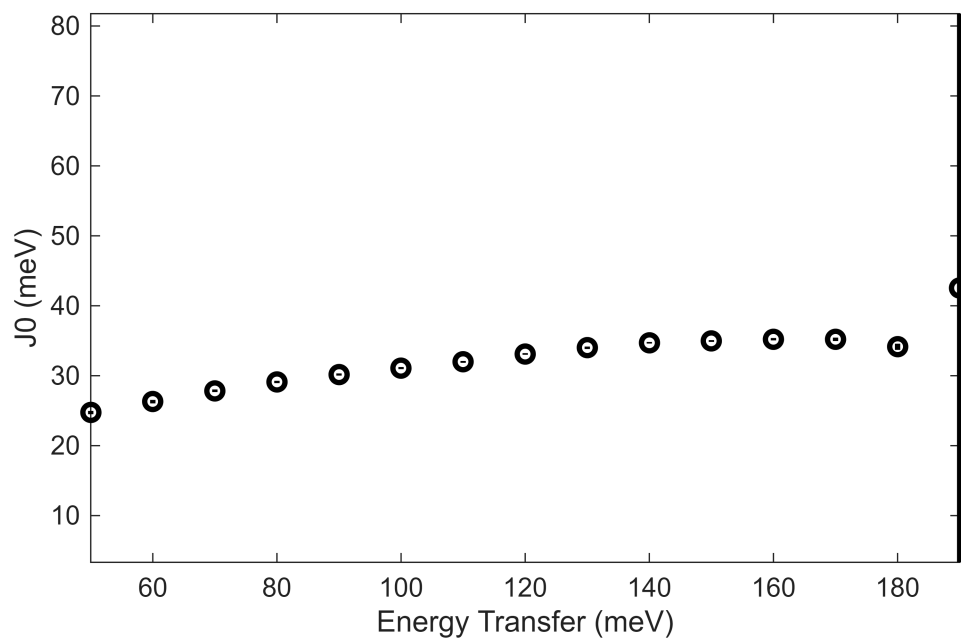
$|\xi| \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $180 \leq E$

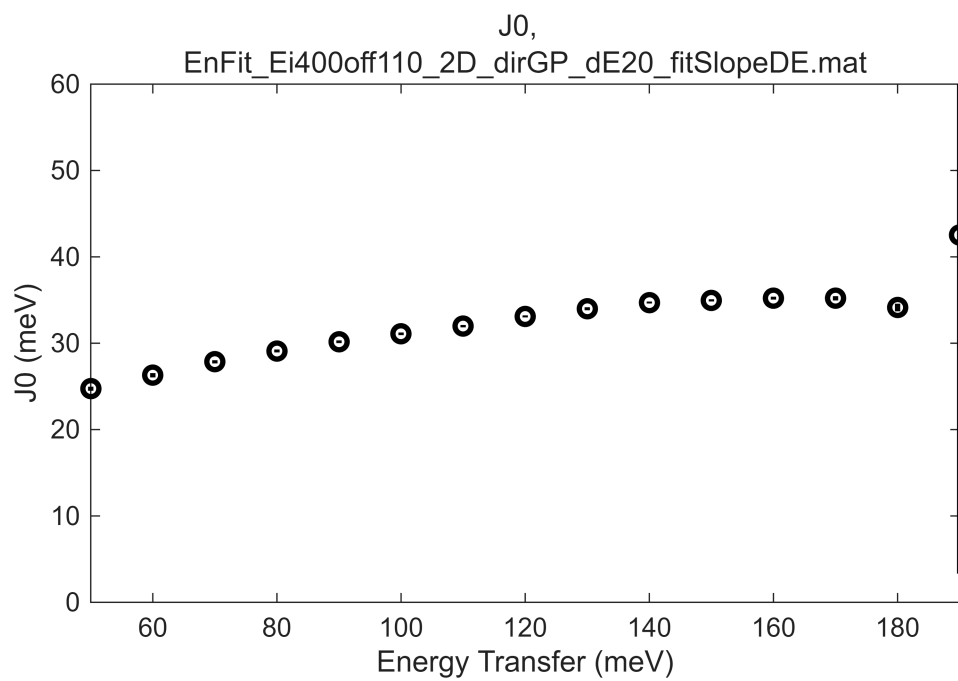
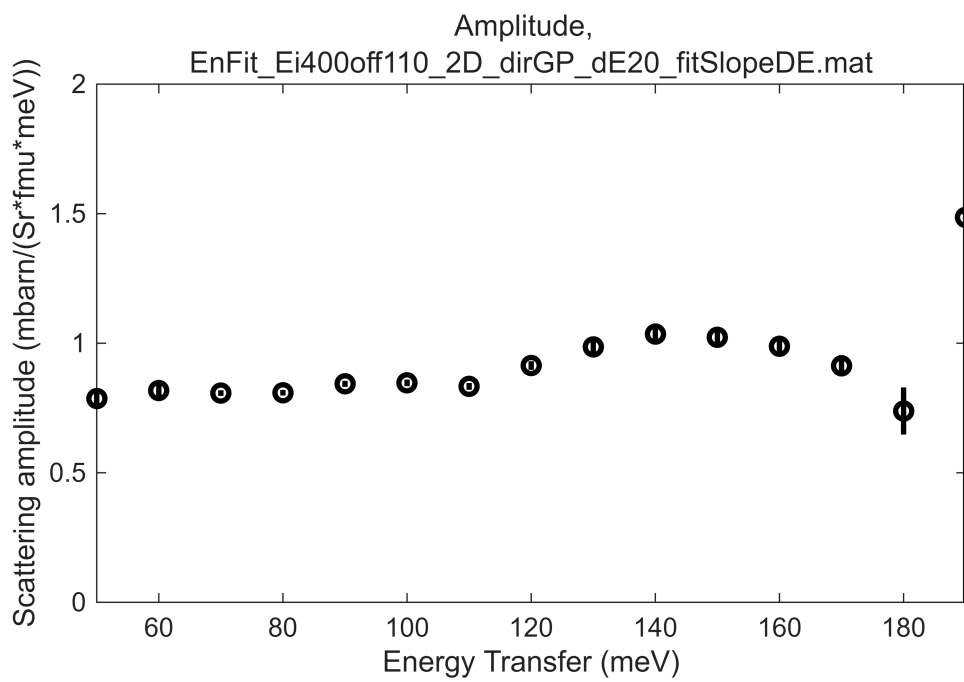
$\zeta = -0.005:0.02:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

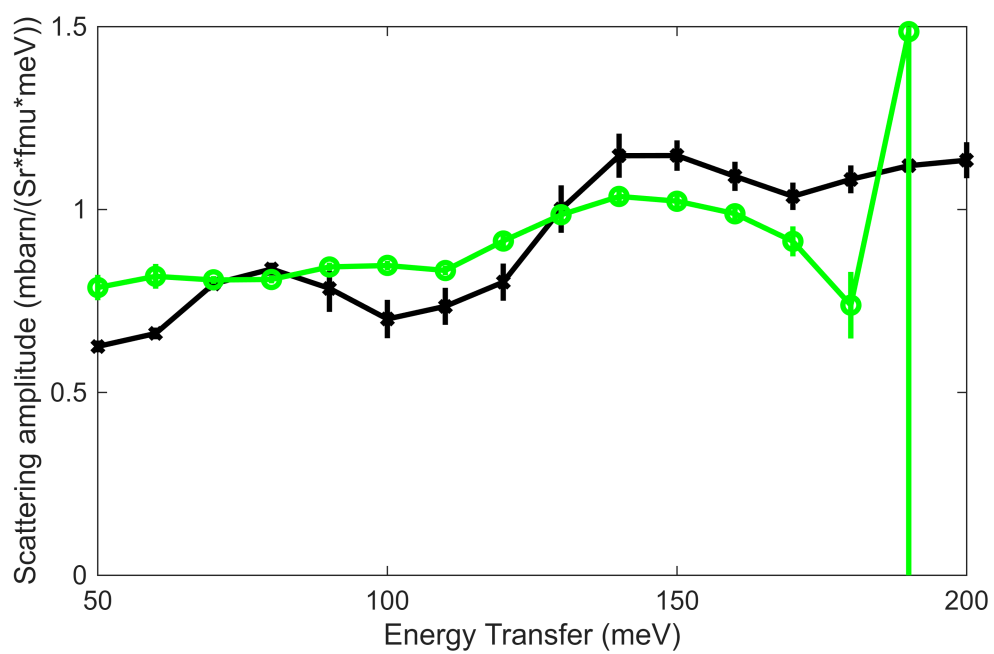
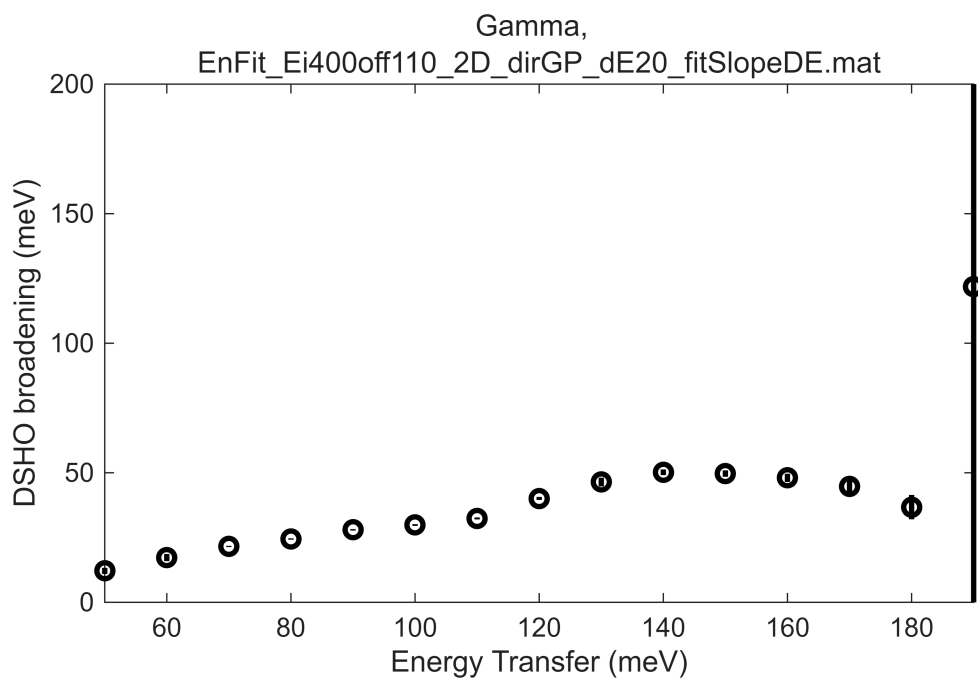


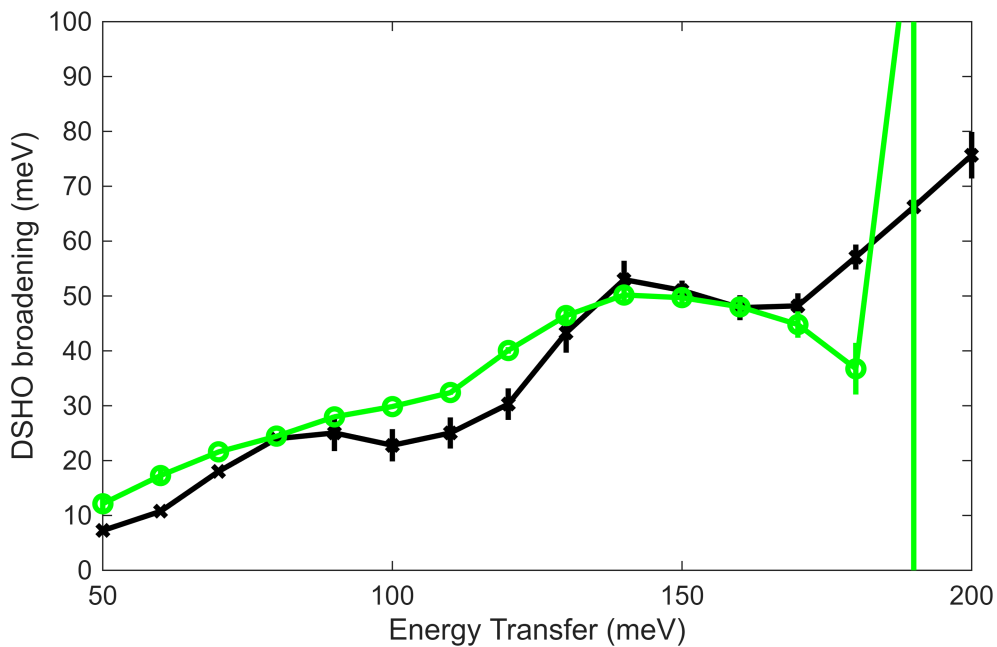
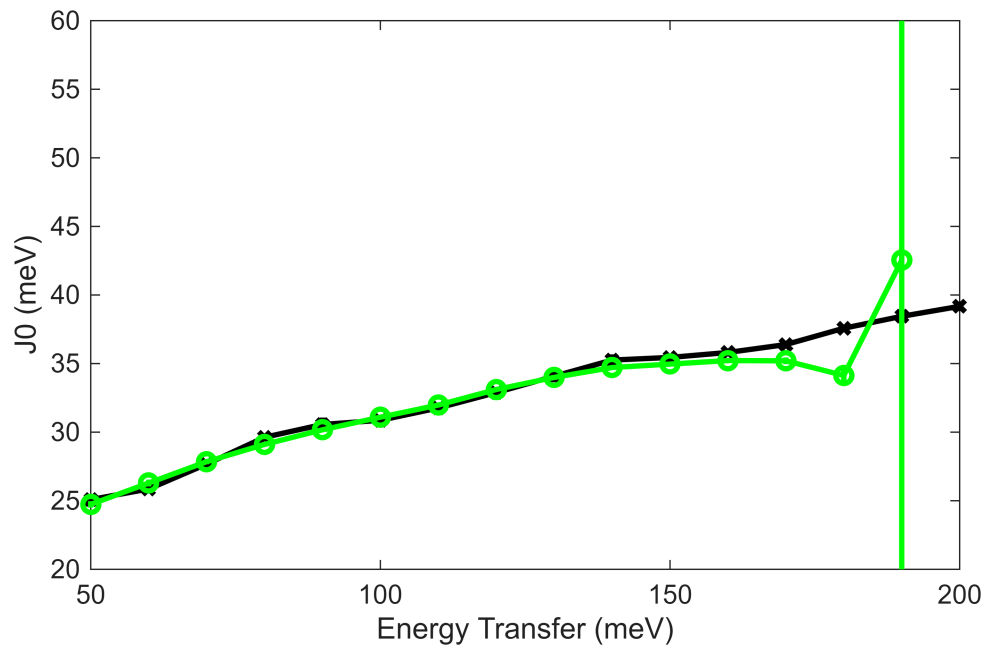
**** En = 200±10











Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Amplitude,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0sym4D_cutsAndFits\EnFit\Ei400off200_2D_dirGP_dE20_fitSlo
peDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
J0,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0sym4D_cutsAndFits\EnFit\Ei400off200_2D_dirGP_dE20_fitSlo
peDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Gamma,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0sym4D_cutsAndFits\EnFit\Ei400off200_2D_dirGP_dE20_fitSlo
peDE.mat

```
%c1,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

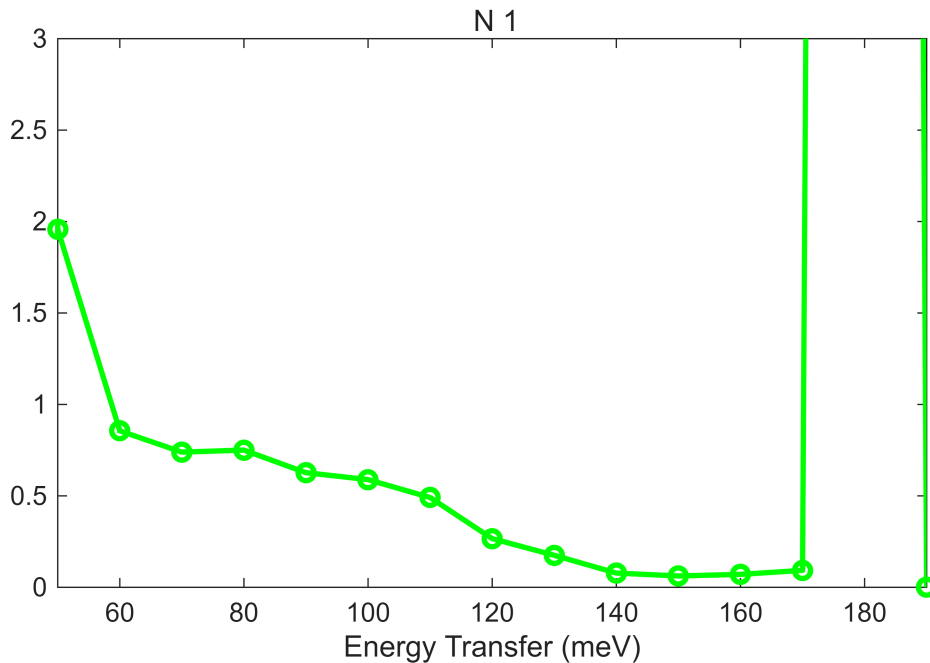
```
fit_res_400 = fit_res_400rec;
```

```
bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
```

```
% original slope parameters
```

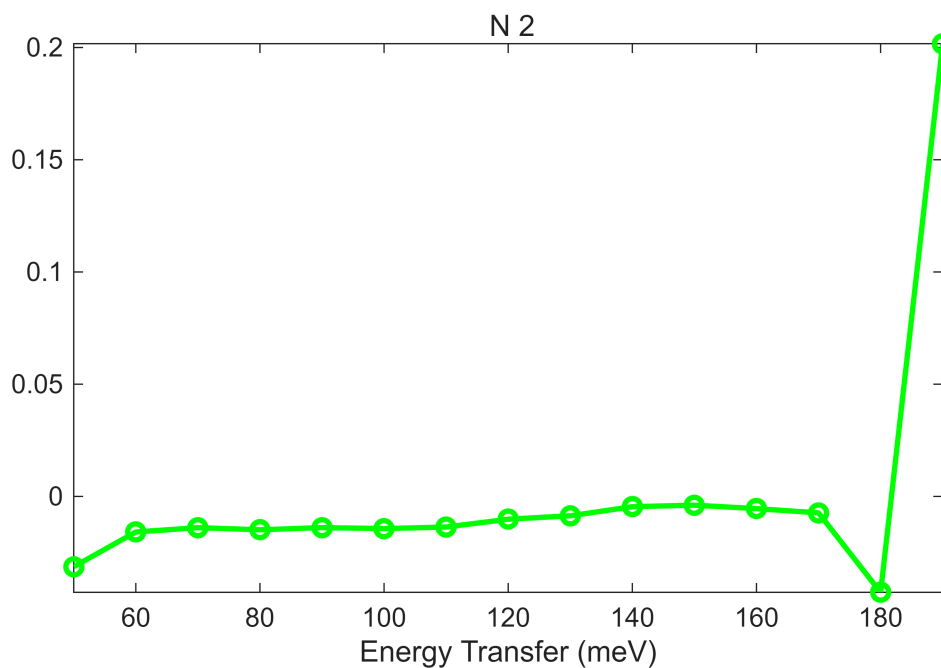
```
pd(bg_data(1));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
```

```
keep_figure; ly 0 3;
```



```
pd(bg_data(2));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
```

```
keep_figure
```



```
fit_resEi400 = fit_res_400rec;
```

```
save(res_name,"fit_resEi400")
```